

Introduction .1

Clickjacking, also known as a "UI redress attack," is a malicious technique used by cybercriminals to trick users into clicking on something different from what they perceive. This is achieved by embedding a hidden frame or iframe on a webpage, which overlays or underlays legitimate content. When users interact with what they believe to be a safe element, they are interacting with the hidden frame, leading to unintended actions such as .financial transactions, data theft, or changes in account settings

This report aims to provide a comprehensive understanding of clickjacking, its methods, impacts, and the countermeasures that can be implemented to protect against it. The report will also include a literature review, the importance of the topic, and recommendations for .both developers and users to mitigate the risks associated with clickjacking

Literature Review .2

Definition and Background 2.1

Clickjacking is a type of cyber attack that exploits the trust users have in the visual interface of a website. The term was first coined by Jeremiah Grossman and Robert Hansen in 2008. The attack involves overlaying a transparent or opaque layer on top of a legitimate webpage, .tricking users into performing actions they did not intend to

Statistics and Cases 2.2

- **Case Study 1:** In 2009, Twitter was vulnerable to a clickjacking attack where users were tricked into clicking a button that automatically tweeted a message without their consent.
- **Case Study 2:** In 2010, Facebook faced a clickjacking attack where users were tricked into "liking" pages they did not intend to, leading to the spread of malicious content.
- **Statistics:** According to a report by Symantec, clickjacking attacks have been on the rise, with a 30% increase in reported cases between 2018 and 2020. The financial impact of these attacks is significant, with losses estimated to be in millions of dollars annually.

Importance of the Topic 2.3

Clickjacking is a serious cybersecurity threat because it exploits the user's trust in the visual interface of a website. Unlike other attacks that rely on technical vulnerabilities, clickjacking exploits human psychology, making it difficult to detect and prevent. The consequences of clickjacking can range from minor annoyances, such as unwanted social media posts, to severe financial losses and data breaches.

Methods of Clickjacking Attacks .3

Iframes 3.1

One of the most common methods used in clickjacking attacks is the use of iframes. An iframe is an HTML element that allows a webpage to be embedded within another webpage. Attackers use iframes to embed legitimate webpages within their malicious sites. When users interact with the visible content, they are actually interacting with the hidden iframe, which may perform unintended actions.

Transparency Manipulation 3.2

Attackers can manipulate the transparency of the iframe to make it invisible or semi-transparent. This makes it difficult for users to detect the hidden elements they are interacting with. For example, an attacker could overlay a transparent iframe on top of a legitimate "Like" button on Facebook, tricking users into liking a page they did not intend to.

Other Techniques 3.3

- **Cursorjacking:** This technique involves manipulating the cursor's position, making users believe they are clicking on one element when they are actually clicking on another.
- **Formjacking:** This is a variation of clickjacking where attackers embed hidden forms within a webpage, tricking users into submitting sensitive information.

4. Impact of Clickjacking

4.1 Financial Transactions

One of the most severe impacts of clickjacking is the potential for financial loss. Attackers can trick users into clicking on buttons that initiate money transfers or purchases, leading to

unauthorized financial transactions.

4.2 Changing Settings

Clickjacking can also be used to alter account or privacy settings. For example, an attacker could trick a user into changing their email settings, allowing the attacker to intercept sensitive communications.

4.3 Data Harvesting

Clickjacking can be used to harvest personal data from users. By tricking users into clicking on hidden elements, attackers can collect sensitive information such as login credentials, credit card numbers, and other personal data.

5. Countermeasures and Protection Methods

5.1 X-Frame-Options

One of the most effective ways to prevent clickjacking is by using the **X-Frame-Options** HTTP header. This header allows web developers to control whether their site can be embedded within an iframe. By setting the X-Frame-Options header to "DENY" or "SAMEORIGIN," developers can prevent their site from being embedded in malicious iframes.

5.2 Content Security Policy (CSP)

The **Content Security Policy (CSP)** is another powerful tool for preventing clickjacking. CSP allows developers to define which sources of content are allowed to be loaded on their site. By configuring CSP to block iframes from untrusted sources, developers can reduce the risk of clickjacking attacks.

5.3 User Awareness

While technical measures are essential, user awareness is also crucial in preventing clickjacking attacks. Users should be educated about the risks of clicking on unknown or suspicious links. They should also be advised to avoid interacting with elements on a webpage that seem out of place or unexpected.

5.4 Browser Extensions

There are several browser extensions available that can help protect against clickjacking. These extensions detect and block malicious iframes, providing an additional layer of security for users.

6. Conclusion

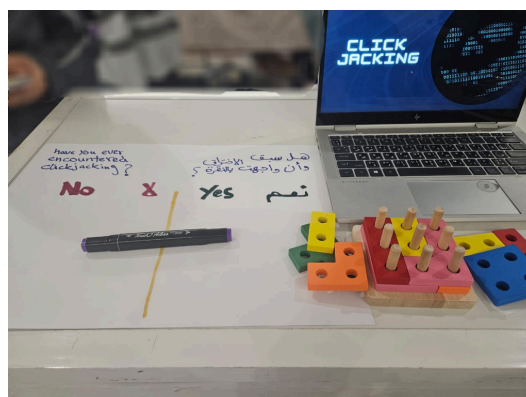
Clickjacking is a serious cybersecurity threat that exploits the trust users have in the visual interface of a website. By understanding how clickjacking works and implementing protective measures such as X-Frame-Options and Content Security Policy, developers can significantly reduce the risk of clickjacking attacks. Additionally, increasing user awareness and educating them about the risks of clicking on unknown or suspicious links can further mitigate the threat.

As cybercriminals continue to develop new and sophisticated methods of attack, it is essential for both developers and users to remain vigilant and proactive in protecting against clickjacking and other cybersecurity threats.

7. References

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8. Booth pictures





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